

Digital Mice: Inducing and Validating Controlled Cognitive Deficits in Frontier Language Models for Mechanism Design

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Abstract

Early-stage mechanism design is a phase-I science: the relevant behavior is human cognition, so designers must run experiments on people before they can learn which mechanisms are robust to bounded rationality. Frontier language models are an appealing in-silico substitute, but a documented failure of LLM “digital twins” is that they are systematically *too* rational and too model-dependent to stand in for the full distribution of human behavior. We turn this distortion into an instrument. Rather than asking a model to reproduce the human population, we start from a capable reasoning substrate and instruct it—through explicit procedural constraints on its deliberation—into specific, controlled cognitive deficits: *digital mice*, the economist’s analogue of the knockout laboratory mouse. Using a canonical second-price sealed-bid auction (SPSB) and a $3 \times 3 \times 3$ panel defined over three cognitive axes (contingent reasoning, forward planning, belief reasoning), we establish four facts. First, the deficits have *construct validity*: engaging each axis moves truthful bidding in the theory-predicted direction, monotonically and with the predicted sign in all five pre-registered directional tests. Second, the panel—augmented by a small set of targeted error strains—attains *calibration coverage* in the sense of support: it can express all five canonical SPSB human-mistake classes above a-priori detection floors. Third, an audit of the models’ own reasoning chains shows the induction is real but *leaky*: a deficit instruction that forbids contingent reasoning still lets the model reconstruct the dominance argument 41% of the time, which both explains and is mitigated by a hardened (“strict”) prompt and worked demonstrations. Fourth, the panel is *externally discriminating*: in a first-price stress test the reasoning strains track the risk-neutral Bayes–Nash benchmark ($b = \frac{2}{3}v$) and overbid relative to

it at the documented human rate, while a strain that overgeneralizes the second-price payment rule bids above value and loses money—a mechanistic winner’s-curse pathology—and four of five reserved strains preserve their behavioral class across base models. Finally, when the panel drives an evolutionary search over a structured space of single-item auctions, the search improves fitness and elevates first-price sealed-bid to the top of the population, agreeing with the a-priori stress test; a transparent re-ranking by scaled deviation from each mechanism’s own optimum tells the complementary story that second-price is easiest to bid in optimally. These results support instructed-deficit language models as a phase-I screening population for mechanism design, to be validated against human experiments rather than treated as a substitute for them.

Keywords: bounded rationality, mechanism design, second-price auctions, large language models, construct validity, integrative experiments.

1 Introduction

Medicine studies humans, but it does not begin with them. Much early drug testing starts with mice, and some of those mice are deliberately engineered to lack a specific protein, receptor, or channel. The deficit is not a nuisance; it is the identification strategy. If a treatment succeeds or fails only when a particular channel is knocked out, the experiment isolates a mechanism that would be entangled in an intact organism. Mechanism design has lacked this luxury. The behavior of interest—contingent reasoning, forward planning, belief formation, strategic sophistication—is bundled in real subjects. A bidder who fails to bid truthfully in a strategy-proof auction may lack contingent reasoning, may fail to plan forward, may hold incorrect beliefs about rivals, or may simply misunderstand the rule. Natural and laboratory experiments rarely deliver a population missing exactly one cognitive module, so the empirical content of theories such as obvious strategy-proofness (Li, 2017), cursed equilibrium (Eyster and Rabin, 2005), and level- k reasoning (Crawford and Iriberri, 2007) is hard to separate.

Large language models have been proposed as simulated economic subjects that could fill this gap, generating experimental data *in silico* before human subjects are recruited (Horton, 2023; Manning et al., 2024). The promise is real, but so is a now well-documented hazard. Peng et al. (2025) show that LLM “digital twins” distort the populations they are meant to mirror along five systematic axes; one of these is *hyper-rationality*—the strongest current reasoning models are, in simple games, often closer to a fully rational agent than

an ordinary human subject—and another is heavy dependence on the particular base model. Faithfully reproducing the entire distribution of human behavior with such a twin is therefore both unattainable and fragile, which counsels against treating an LLM as a drop-in replica of a human population.

Our approach turns hyper-rationality from a bug into the instrument. We do not ask a model to match the human distribution. Instead we start from the capable, hyper-rational substrate and instruct it to “play dumb” in a controlled way—reason only myopically, enumerate only local contingencies, ignore second-order beliefs—through explicit procedural constraints on its deliberation, not through dependence on hidden quirks of one model. This yields *digital mice*: language-model agents with specific, engineered cognitive deficits. The deficit is induced by rationing which reasoning steps the model may use, while leaving its objective (maximize profit) intact, exactly as a knockout removes one faculty while leaving the organism otherwise whole.

Research question. Can a frontier reasoning model be instructed into stable, separable bounded-rationality states that (i) produce the behavioral signatures predicted by cognitive theory and (ii) span the empirically documented space of human mistakes, in a canonical second-price sealed-bid auction—and can such a panel then screen candidate mechanisms for robustness to bounded rationality?

An integrative-experiment design. Rather than run a single experiment, we define a *space* and sample it, in the spirit of integrative experiment design (Almaatouq et al., 2024; Alsobay et al., 2025). There are two spaces. The first is cognitive: a $3 \times 3 \times 3$ panel that crosses three cognitive axes at three levels each, yielding 27 strains. The second is institutional: a structured genotype space of single-item auctions over which an evolutionary search can move. Figure 1 lays out the full pipeline—from a base model, to instructed deficits, to auction environments, to behavioral measurement, to a panel-based fitness that drives mechanism search and, finally, to human validation.

Contribution. We answer the question affirmatively for the SPSB environment and demonstrate the screening loop. (1) *Construct validity*. Over the 27-strain panel (810 bidder-decisions), engaging each cognitive axis raises the truthful-bid rate by +0.23 (contingent reasoning), +0.14 (forward planning), and +0.05 (belief reasoning); all five pre-registered directional hypotheses pass their sign test. (2) *Calibration coverage*. Augmented by two targeted error strains, the panel expresses all five canonical SPSB mistake classes above their detection floors. (3) *Reasoning audit*. Parsing the models’ logged reasoning

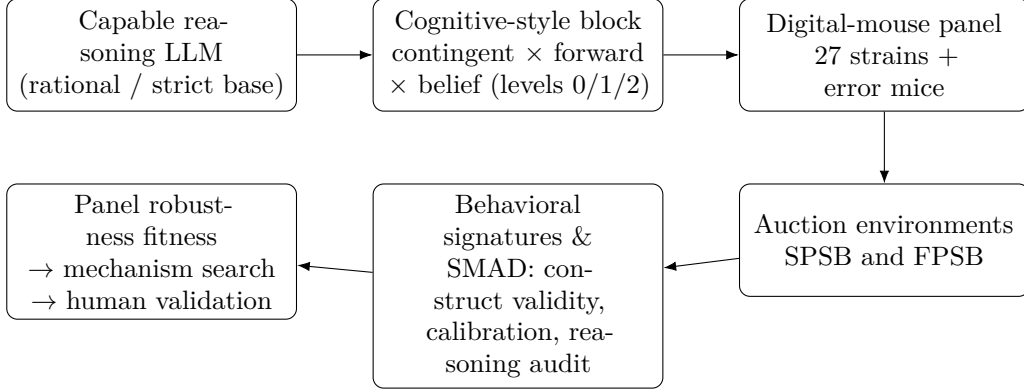


Figure 1: The digital-mouse pipeline. A capable reasoning model is instructed into a panel of controlled cognitive deficits (top row); the panel is run in auction environments and its behavior is measured against theory (bottom right); the resulting panel-based robustness objective drives a search over auction mechanisms, whose output is a short list of candidate mechanisms for human-subject validation (bottom left).

shows the deficit instruction works in the right direction but leaks: contingent-off strains still reconstruct the dominance argument 41% of the time (versus 77% when contingent reasoning is engaged), which motivates a hardened prompt and worked demonstrations. (4) *External validity*. In a first-price stress test the reasoning strains shade to the risk-neutral BNE ($b = \frac{2}{3}v$, median ratios 0.65–0.67) and overbid relative to it at 38%, while a second-price-overgeneralizer mouse bids above value and loses money; four of five reserved strains preserve their behavioral class across base models. (5) *Mechanism search*. An evolutionary search that scores candidates on the panel independently elevates first-price sealed-bid to the top of the population—the same mechanism the a-priori stress test selected—and a transparent re-ranking by deviation from each mechanism’s own optimum shows that second-price, by contrast, is the easiest mechanism to bid in optimally. We are explicit about a boundary: the clean procedural basis naturally produces truthful and underbidding behavior but not overbidding, which we recover with a small number of hand-specified heuristic strains, and we treat the panel as phase-I screening evidence rather than a prediction of human population shares.

Roadmap. Section 2 constructs the digital-mouse panel. Section 3 gives the experimental design and the outcome variables, including the deviation metric (SMAD) and the panel-based robustness objective. Sections 4 and 5 report construct validity and calibration coverage. Section 6 audits the reasoning chains, and Section 7 shows that worked demonstrations sharpen induction. Section 8 is the external-validity chapter: the first-price stress test, a map from deficits to human error theories, and cross-model

portability. Section 9 drives an evolutionary mechanism search with the panel. Section 10 discusses limits (including the digital-twin distortions), Section 11 lays out a path to human validation, and Section 12 concludes. Appendix A gives the full prompts; Appendix B specifies a planned outcome-instruction robustness check.

2 The Digital-Mouse Panel

2.1 Construction: a rational base plus a deficit instruction

A digital mouse is a single prompt assembled from two layers. The first layer, the *base*, is identical across all strains in a variant and fixes the objective. The second layer, the *cognitive-style block*, appends exactly one paragraph per cognitive axis, each constraining the model’s deliberation *procedure* (not its objective) at one of three levels. The intervention is therefore a reasoning-procedure control, analogous to knocking out one faculty while leaving the organism’s goal intact.

Two base layers, and why the strict one is the workhorse. The natural first base simply makes profit the top priority:

*Your TOP PRIORITY is to place bids which maximize your profit in the long run.
Learn from the history of previous rounds in order to maximize your total profit.
Don’t forget the values are redrawn independently each round.*

This *standard* base has a defect that matters for our purpose: it leaves room for a very capable model to “see through” an assigned deficit and revert to the optimal strategy, because the instruction it is told to prioritize (maximize profit) points straight at the dominant strategy. We document this leakage directly in Section 6. We therefore also build a *strict* base that inverts the priority and explicitly forbids the dominance shortcut:

You are acting as a digital mouse with a specified cognitive procedure. Your TOP PRIORITY is to follow the cognitive procedure exactly... if the cognitive procedure conflicts with the profit-maximizing strategy, the cognitive procedure wins. ... Do not invoke auction theorems, dominant-strategy labels, or Vickrey truth-telling shortcuts unless the allowed procedure below gives you enough reasoning steps to derive them.

The strict base is the workhorse for any analysis that must hold a deficit fixed against a capable model—the cross-model and evolutionary-search results of Sections 8 and 9. The construct-validity and calibration results of Sections 4–5 use the standard base, which

we report transparently; the reasoning audit (Section 6) is precisely the evidence that motivates migrating the whole battery to the strict base, a re-run we flag rather than assume.

The three axes. Each axis paragraph ranges from a level-0 ablation that forbids a class of reasoning to a level-2 paragraph that mandates it. For the contingent-reasoning axis:

Contingent reasoning, level 0 (ablated). *Do NOT enumerate possible opponent bids, best responses, dominance cases, or theorems about opponent-independent optimal bids. Treat the auction as a fixed environment and choose from a simple own-value rule.*

Contingent reasoning, level 2 (engaged). *You may compare candidate bids across the full range of opponent bids, including worst-case and dominance-style reasoning. If one bid is robust across all cases, use it.*

The forward-planning axis runs from “do not trace consequences” (a fixed margin-of-safety underbid) to a multi-step decision tree, and the belief-reasoning axis from “do not model others” to explicit second-order beliefs; the full text is in Appendix A. A strain is the base plus a triple $(c, f, b) \in \{0, 1, 2\}^3$. The all-ablated triple $(0, 0, 0)$ is the most deficient mouse; the all-engaged triple $(2, 2, 2)$ is closest to the rational base. To avoid overloading the word “repair,” we reserve it for its prompt-level meaning—a capable model spontaneously *repairing* its own assigned deficit, the failure mode of Section 6—and use “engage” for the analyst’s act of moving an axis from level 0 to level 2.

2.2 The $3 \times 3 \times 3$ grid

Varying three axes at three levels each gives the 27-strain panel of Table 1. The base supplies the *goal* (maximize profit) and the cognitive-style block rations the *reasoning steps* the mouse may use to pursue it. A mouse told not to enumerate opponents, not to trace payoffs, and not to model others must fall back on a crude own-value heuristic; a mouse granted all three faculties can in principle reconstruct the dominant strategy—which, in this second-price environment, is truthful bidding. (We are careful with this claim: a dominant strategy exists *here* because the mechanism is strategy-proof; in the first-price environment of Section 8 there is no dominant strategy, only a Bayes–Nash best response, and we phrase the benchmark accordingly.)

Table 1: The $3 \times 3 \times 3$ digital-mouse panel. Each strain is one combination of axis levels; level 0 is the engineered (ablated) deficit, level 2 the engaged faculty.

Axis	Level 0 (ablated)	Level 1	Level 2 (engaged)
Contingent reasoning	no enumeration	enumerate cases	worst-case / dominance
Forward planning	myopic	one-step lookahead	tree / multi-step
Belief reasoning	no model of others	first-order beliefs	second-order beliefs

3 Experimental Design and Methodology

Environment. The mechanism is a single-item second-price sealed-bid auction with $N = 3$ bidders. Private values are drawn independently and uniformly on $[0, 50]$; each bidder observes only its own value and submits one sealed bid on a \$0.1 grid. Under standard preferences truthful bidding is a weakly dominant strategy, which makes truthfulness a clean construct-validity target: a more capable mouse should bid closer to its value.

Strains and runs. We instantiate all 27 basis strains with the same base model (`gpt-5.4-mini`, temperature 0.5) and collect 30 bidder-decisions per strain (810 in total). Because the clean procedural basis does not naturally generate *overbidding* (explained next), we additionally author a small set of targeted *error mice*.

How the error mice are built, and why they are needed. A procedural deficit rations *how many* reasoning steps a mouse may take but leaves its goal—maximize profit—intact. Rationing steps can only make a profit-seeking bidder *more cautious*: it produces truthful bidding (when enough steps remain to find the dominant strategy) or underbidding (a margin-of-safety fallback), but it has no route to systematic *overbidding*, which requires a positively wrong belief rather than a missing reasoning step. Overbidding is, however, a first-order regularity in human auctions (Kagel and Levin, 1993). We therefore author error mice as small heuristic strains, each expressing one documented content error. Two have completed runs and feed the scorecard below. The “second-price overgeneralizer” is built from a single mistaken shortcut:

You strongly focus on the fact that the winner pays the second-highest bid, not their own bid. Overgeneralize this rule: treat bidding above your value as mostly safe because you expect not to pay your own bid.

and the “payment panic” mouse from a fear of its own bid:

You are afraid that your bid is the amount you might have to pay... choose the smallest positive bid allowed by the auction grid... Even if your value is high, protect yourself by bidding 0.1 rather than trying to win.

Both use the strict framing (“follow the cognitive style even if a more careful strategy is available”). The distinction between the procedural basis and the authored error mice is an honest construct-validity boundary, not a patch: the basis cleanly demonstrates that *rationing reasoning* reproduces the cautious-mistake half of the space, and the error mice show that the *remaining* half requires injecting a specific wrong belief.

Behavioral classes. For each bidder-decision we record the bid b , the value v , and the deviation $b - v$. A truthful bid is $|b - v| \leq 0.05$; moderate underbidding is $b < v - 0.05$ (excluding near-zero bids); overbidding is $b > v + 0.05$; extreme overbidding is $b - v \geq 5$; near-zero bidding is $b \leq 0.1$ with $v > 5$. These operationalize the five canonical SPSB mistake classes scored in Section 5.

3.1 Outcome variables, a deviation metric, and the robustness objective

We use two kinds of summary. The first, SMAD, is a single transparent scalar that answers “how close do bidders come to this mechanism’s own optimum?” The second, F_{robust} , is a composite that the evolutionary search of Section 9 optimizes. We introduce both here and use SMAD as the interpretable lens on the composite later.

SMAD: scaled deviation from a mechanism’s own optimum. Many mechanisms declare a benchmark bid $b^*(v)$ that a correctly reasoning bidder should submit: $b^* = v$ for second-price (truthful), $b^* = \frac{n-1}{n}v$ for first-price, and so on. Define

$$\text{SMAD} = 100 \cdot \frac{\text{mean}_a |b_a - b^*(v_a)|}{\text{mean}_a |b^*(v_a)|}, \quad (1)$$

the mean absolute deviation of realized bids from the mechanism’s own optimum, expressed as a percentage of the mean optimal bid. Lower is better; SMAD is undefined for mechanisms with no scalar benchmark (e.g. posted-price binary-accept). Unlike the composite below, SMAD lives on one interpretable scale, which is why we use it in Section 9 as a cross-check.

Per-auction quantities, and why we keep them separate. Fix a mechanism and a persona panel $\mathcal{P} = \{p_1, \dots, p_M\}$ and run R independent auctions per persona. Index completed auctions by $a \in \mathcal{A}$, each with $N = 3$ bidders, values $v_{a,i}$, bids $b_{a,i}$, winner $i^*(a) = \arg \max_i b_{a,i}$, and payment $\rho(a)$ set by the pricing rule. With $v_a^{\max} = \max_i v_{a,i}$,

$$\text{eff}(a) = \frac{v_{a,i^*(a)}}{v_a^{\max}}, \quad s(a) = \text{std}_{i: v_{a,i} > 0} \left(\frac{b_{a,i}}{v_{a,i}} \right), \quad \pi(a) = v_{a,i^*(a)} - \rho(a). \quad (2)$$

We deliberately keep these three as separate components rather than collapsing them into one number: efficiency measures whether the item is allocated to the highest-value bidder, the bid/value dispersion s measures whether bidders coordinate on a common “obvious” shading (an OSP-style desideratum), and winner surplus π measures tail risk borne by bidders. A single scalar would conflate three economically distinct objectives; SMAD is the one place we do want a single number, and it answers a different question (closeness to the declared optimum) than any of the three.

Panel aggregates and lower tail. With $A = |\mathcal{A}|$, value scale $\bar{v} = \max_a v_a^{\max}$, and failure rate $\phi = F/A_{\text{att}}$ (responses that fail to parse), define normalized revenue $\hat{\rho} = \frac{1}{A\bar{v}} \sum_a \rho(a)$, mean efficiency $\overline{\text{eff}}$, a simplicity proxy $\sigma = 1 - \frac{1}{A} \sum_a s(a)$, and normalized surplus variance $\widehat{\text{Var}} = \text{Var}_a(\pi(a))/\bar{v}^2$. Robustness rewards good *worst-case* behavior across the panel, so we also take the per-persona minima and a low quantile:

$$E_{\min} = \min_p \overline{\text{eff}}_p, \quad E_{20} = Q_{0.2}(\{\overline{\text{eff}}_p\}), \quad \Sigma_{\min} = \min_p \sigma_p, \quad \widehat{V}_{\max} = \frac{\max_p \text{Var}_p(\pi)}{\bar{v}^2}. \quad (3)$$

Intended-action rate. Each mechanism declares the action a correctly reasoning bidder should take, $\tau \in \{\text{TRUTHFUL}, \text{SHADE}, \text{AGGRESSIVE}\}$, with tolerance $\epsilon = 0.1$; for a scoreable (v, b) with $v > 0$,

$$m_\tau(v, b) = \begin{cases} \mathbf{1}[|b - v| \leq \epsilon] & \tau = \text{TRUTHFUL}, \\ \mathbf{1}[\kappa < b < v - \epsilon] & \tau = \text{SHADE}, \\ \mathbf{1}[b + \epsilon \geq \alpha v] & \tau = \text{AGGRESSIVE}, \end{cases} \quad (4)$$

with floor κ and $\alpha = 0.4$; μ_τ is the mean of m_τ over scoreable decisions.

Fitness. The aggregate and robust fitness are weighted sums

$$F_{\text{agg}} = w_1\hat{\rho} + w_2\overline{\text{eff}} + w_3\mu_\tau + w_4\sigma - w_5\widehat{\text{Var}} - w_6\phi, \quad (5)$$

$$F_{\text{tail}} = w_7E_{\text{min}} + w_8E_{20} + w_9\Sigma_{\text{min}} - w_{10}\widehat{V}_{\text{max}}, \quad (6)$$

$$F_{\text{robust}} = F_{\text{agg}} + F_{\text{tail}}, \quad (7)$$

with defaults $(w_1, \dots, w_{10}) = (1, 1, 0.5, 0.5, 0.25, 1, 0.5, 0.5, 0.25, 0.25)$ grouping into domain performance (w_1, w_2) , intended action (w_3) , simplicity (w_4, w_9) , regret (w_5, w_{10}) , compliance (w_6) , and cognitive fragility (w_7, w_8) . The sign convention is that fragility, regret, and failure enter negatively, so a mechanism scores well only if it is efficient, induces its intended action, and does so *uniformly* across the panel. We acknowledge that F_{robust} is a deliberately multi-objective composite and hence harder to read than a single number; this is exactly why Section 9 re-ranks the evolved mechanisms by SMAD (1) as a transparent check.

4 Result 1: Construct Validity

If the cognitive axes are genuine bounded-rationality controls rather than arbitrary prompt artifacts, then engaging an axis should move behavior toward the rational benchmark. Figure 2 plots the truthful-bid rate at each axis level, marginalizing over the other two. All three axes are monotone or near-monotone in the engaging direction, and contingent reasoning produces the largest swing.

Table 2 reports the five pre-registered directional tests. For each axis we test that engaging it raises the truthful rate (expected positive) and, where applicable, lowers the underbidding rate (expected negative). All five pass: the realized sign matches the prediction and exceeds the minimum-magnitude bar in every case. The directional structure is exactly what theory predicts—more contingent reasoning and more forward planning push the agent toward its dominant strategy—which is the core construct-validity claim.

Figure 3 shows that the strains are also *separable* at the level of raw behavior. The rational mouse (contingent reasoning engaged) bids on the 45° truthful line; the all-ablated mouse systematically underbids; and the targeted error mice occupy distinct regions—persistent overbidding above the line and near-zero bidding along the floor. The mean absolute deviation falls from 2.78 for the most deficient basis strain to essentially zero (0.003) for the most rational one.

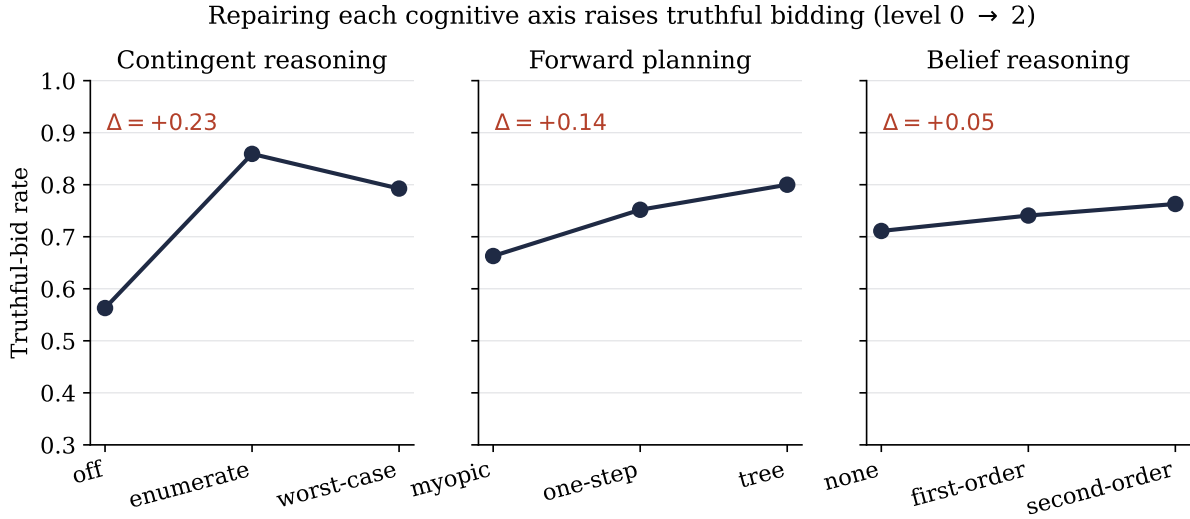


Figure 2: Construct-validity marginals. Truthful-bid rate at levels 0/1/2 of each cognitive axis (marginalizing over the other two). Δ is the level-2-minus-level-0 change. Engaging each axis raises truthfulness, with the largest effect on the contingent-reasoning axis.

Table 2: Construct-validity diagnostics: five pre-registered directional tests on the cognitive axes. Δ is the level-2-minus-level-0 change in the metric. All tests pass.

Axis	Metric	Δ (L2–L0)	$ \Delta $ bar	Expected sign	Pass
Contingent reasoning	truthful rate	+0.230	0.05	positive	✓
Contingent reasoning	underbid rate	−0.230	0.05	negative	✓
Forward planning	truthful rate	+0.137	0.05	positive	✓
Forward planning	underbid rate	−0.137	0.05	negative	✓
Belief reasoning	truthful rate	+0.052	0.03	positive	✓

Notes. Sign rate = 1.00 (5 of 5). Underlying level means (truthful rate): contingent 0.56 $\rightarrow$ 0.86 $\rightarrow$ 0.79; forward 0.66 $\rightarrow$ 0.75 $\rightarrow$ 0.80; beliefs 0.71 $\rightarrow$ 0.74 $\rightarrow$ 0.76. Standard base model; 810 bidder-decisions.

5 Result 2: Calibration as Support Coverage

Construct validity establishes that the axes do what theory says. Calibration asks a different question, and it is important to be precise about *which* question. We are **not** doing moment matching: we do not claim that the panel reproduces the human *share* of each mistake. We are testing *support coverage*—can the panel *express* each mistake class at all? This is the relevant question for a screening instrument: a panel that cannot produce overbidding can never stress-test a mechanism against overbidders, regardless of the human rate.

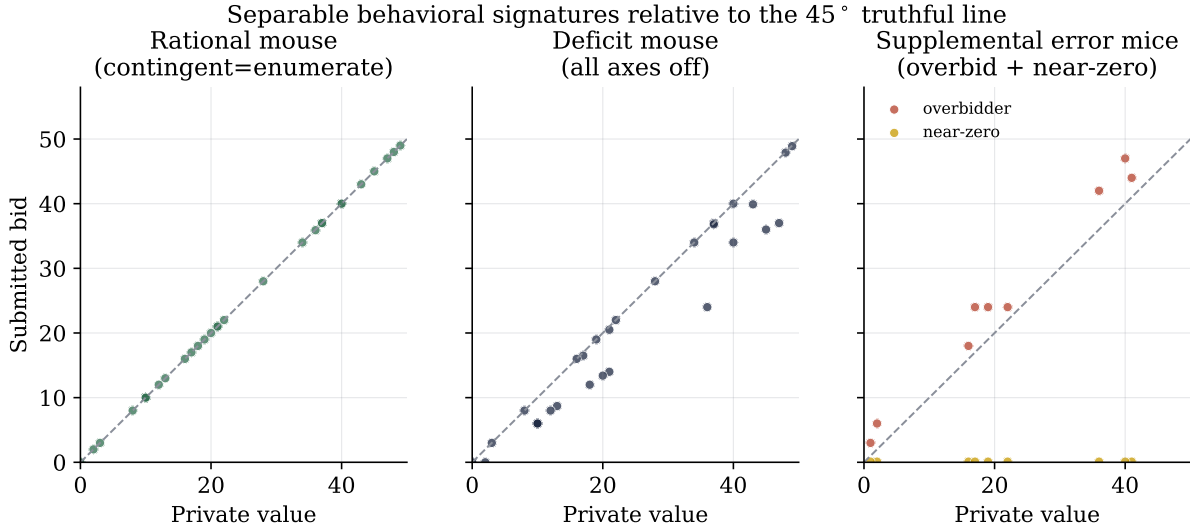


Figure 3: Separable behavioral signatures. Bid versus private value for a rational basis strain, the all-ablated strain, and the two supplemental error mice. The dashed line is the 45° truthful benchmark.

Where the thresholds come from. For each class we set an a-priori *detection floor*—the rate above which we are willing to say “the panel can express this behavior”—and ask whether some strain clears it. These floors are fixed in advance and are deliberately *not* human population shares: truthful 0.50, moderate underbid 0.10, overbid 0.10, extreme overbid 0.05, near-zero 0.05. They encode a detectability bar (a class counts as covered if a strain produces it at least one-tenth to one-half of the time), not a calibration target. Reading the bars in Figure 4 as “distance above the detection floor” rather than “distance from the human share” is the whole point.

Table 3 reports the scorecard. All five classes are covered. Truthful and overbidding behavior are expressed at very high rates (0.97 and 1.00); moderate underbidding, extreme overbidding, and near-zero bidding at intermediate rates (0.63, 0.44, 0.78). Crucially, the table makes explicit which classes the procedural construction reaches on its own (truthfulness and underbidding, via basis strains) and which require the hand-specified error mice (overbidding and near-zero, via the two authored strains).

6 Result 3: Auditing the Reasoning Chains

Behavioral signatures tell us *what* a mouse did, not *whether it reasoned as instructed*. Because the models log their plan (a free-text chain-of-thought) before bidding, we can read the reasoning directly and ask: does a strain that is forbidden to reason contingently

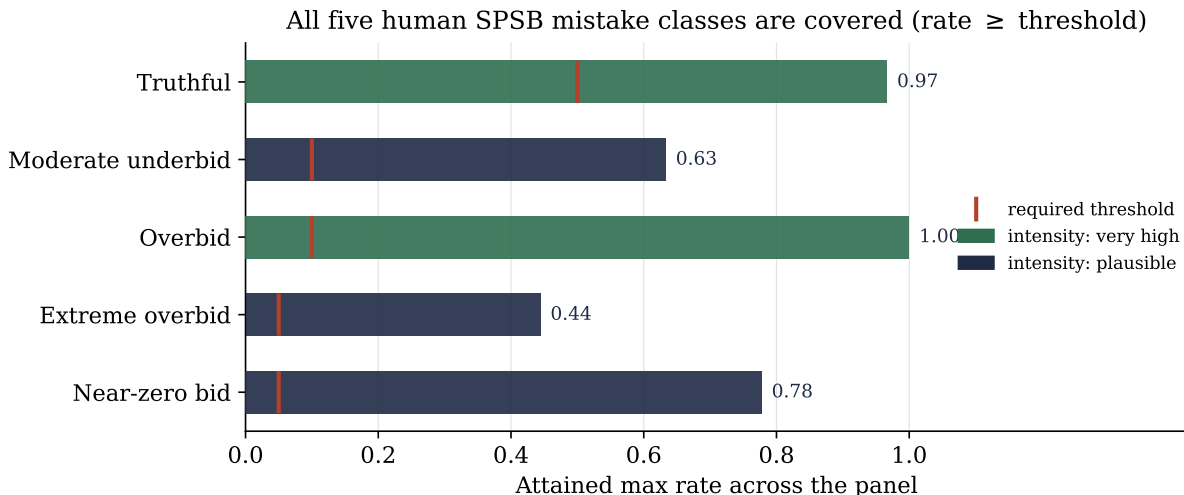


Figure 4: Calibration coverage. For each human-mistake class, the maximum rate attained by any strain (bar) against its a-priori detection floor (red marker). Bar color encodes the intensity band. All five classes clear their floor. The floors are detectability bars, not human population shares.

actually refrain from doing so? We parse the 810 logged plans and flag each one for whether it invokes the second-price dominance/truthful argument—the very move a contingent-reasoning ablation ($c = 0$) is instructed not to make. (We tightened the classifier to catch genuine derivations of the dominant strategy while excluding boilerplate restatements of the payment rule.)

Figure 5 and the underlying rates tell a two-part story. First, the instruction works *in the right direction*: contingent-ablated strains invoke the dominance argument far less often than strains where the axis is engaged (mean invocation 0.41 at $c=0$ versus 0.77 at $c=1$ and 0.66 at $c=2$). Second, the instruction is *leaky*: a 0.41 invocation rate means that even when told not to reason contingently, the capable model still reconstructs the dominance argument in roughly two of every five plans. A representative leak, from a $c=0$ strain, reads: “In a second-price auction, truthful bidding is the standard dominant strategy because it avoids overpaying while preserving the chance to win.” The model is doing exactly what the deficit forbids.

This audit does two jobs. It is independent evidence that the induction is real (ablation moves the reasoning, not just the output), and it diagnoses precisely why the standard base is fragile against a strong model: the dominance argument leaks back in through the very objective the standard base prioritizes. The leak is heaviest for strains that engage the other two axes while ablating contingent reasoning (e.g. `c0_f2_b2`, invocation 0.53), and—tying the threads together—that same strain is the one whose deficit a stronger

Table 3: Support-coverage scorecard for the five canonical SPSB mistake classes. “Max rate” is the highest rate attained by any strain; “best strain” attains it. “Floor” is the a-priori detection bar, not a human share. All classes are covered.

Mistake class	Metric	Floor	Max rate	Best strain	Intensity	Covered
Truthful	truthful rate	0.50	0.967	c1_f0_b0	very high	✓
Moderate underbid	moderate underbid rate	0.10	0.633	c0_f0_b0	plausible	✓
Overbid	overbid rate	0.10	1.000	overgeneralizer [†]	very high	✓
Extreme overbid	extreme overbid rate	0.05	0.444	overgeneralizer [†]	plausible	✓
Near-zero bid	near-zero bid rate	0.05	0.778	payment panic [†]	plausible	✓

Notes. Coverage rate = 1.00 (5 of 5). [†] Supplemental error strain (spsb_error_second_price_overgeneralizer, spsb_error_payment_panic_near_zero); all other best strains are members of the clean 27-cell procedural basis.

model repairs away in the cross-model test of Section 8. The remedy is to make the deficit harder to override, which motivates both the strict base (Section 2) and the worked demonstrations we turn to next.

7 Result 4: Demonstrations Sharpen Induction

The panel so far induces a deficit by *instruction*. A natural alternative, familiar from few-shot chain-of-thought prompting, is to also *demonstrate* the bounded procedure. Concretely, we append to a persona one fully worked example placed after the cognitive-style block and before the live decision: it states a sample private value, walks through only the reasoning the strain is allowed to use, and ends in a bid—a margin-of-safety underbid for the all-ablated strain, and the contingent/forward/belief reasoning that yields a truthful Vickrey bid for the all-engaged strain. The full demonstration text is in Appendix A. We then compare instruction-only and demonstration-augmented personas on *identical* second-price value draws.

Table 4 shows that demonstrations make induction both sharper and more consistent. For the all-ablated strain, the worked example pulls the mean bid/value ratio down toward the demonstrated ≈ 0.7 , eliminates truthful bidding entirely (truthful rate $0.39 \rightarrow 0.00$), and *lowers* the dispersion of bid/value ratios ($\text{sd } 0.124 \rightarrow 0.081$). For the all-engaged strain the truthful demonstration produces exact Vickrey truth-telling on every decision (truthful rate $0.83 \rightarrow 1.00$, dispersion $0.076 \rightarrow 0.000$). Demonstration is thus a stronger control than instruction: it lowers behavioral variance and makes the mouse follow the prescribed procedure more faithfully.

This result also reframes a pilot failure. In an early cross-model pilot the stronger model

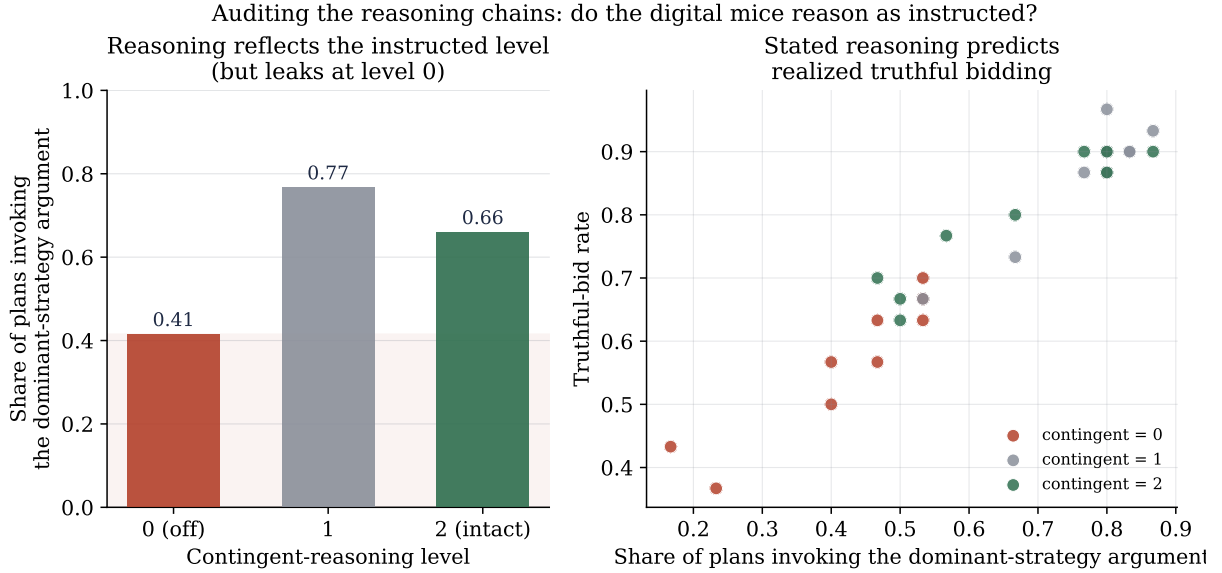


Figure 5: Auditing the reasoning chains. Left: rate at which a strain’s logged plans invoke the second-price dominance argument, by contingent-reasoning level; ablation ($c=0$) lowers invocation but does not eliminate it. Right: dominance-invocation rate versus truthful-bid rate across the 27 strains, colored by contingent level; plans that reconstruct the dominance argument bid truthfully more often, and the ablated strains (red) cluster low on both.

spontaneously repaired the most deficient strain into truthful bidding—which we initially read as a portability failure but now read as a *prompt-design* failure: an instruction-only deficit, on the standard base, is simply too weak to hold against a capable model (Section 6 quantifies the leak). The demonstration result suggests the fix is to induce the deficit by worked CoT rather than instruction alone. Carrying the entire panel over to a CoT-induced, strict-base construction is the natural next step; because it requires re-running the model battery, we flag it as a deliberate (cost-bearing) extension rather than fold it in here.

8 External Validity: A First-Price Stress Test and Cross-Model Portability

A panel is useful for screening only if it reproduces known mechanism regularities and if its verdicts are not an artifact of the base model. This section tests both, and maps the panel’s behaviors onto the human-error theories they are meant to stand in for.

Table 4: Instruction-only versus chain-of-thought (CoT) demonstration, on identical second-price draws ($n = 18$ bidder-decisions per cell). Demonstration shifts behavior toward the demonstrated procedure and lowers the dispersion of bid/value ratios.

Strain	Construction	mean b/v	truthful rate	sd(b/v)
c0_f0_b0 (all-ablated)	instruction only	0.89	0.39	0.124
c0_f0_b0 (all-ablated)	+ CoT demonstration	0.77	0.00	0.081
c2_f2_b2 (all-engaged)	instruction only	0.98	0.83	0.076
c2_f2_b2 (all-engaged)	+ CoT demonstration	1.00	1.00	0.000

8.1 First-price stress test

We move from the second-price environment to a *first-price* sealed-bid auction, where theory and human data are sharp: with $n = 3$ risk-neutral bidders and uniform values, the symmetric Bayes–Nash equilibrium is to shade to $b = \frac{n-1}{n}v = \frac{2}{3}v$, and the canonical experimental regularity is overbidding *relative* to this risk-neutral benchmark (Kagel and Levin, 1993). Note there is no dominant strategy here—only a best response in beliefs—so the benchmark is BNE, not dominance.

Figure 6 and Table 5 report the panel’s first-price behavior. The three reasoning strains track equilibrium: median bid/value ratios are 0.65–0.67, on the $\frac{2}{3}$ BNE line, and across all reasoning decisions the mean ratio is 0.67 with 38% of bids above the risk-neutral benchmark—the documented overbidding regularity. The two error mice separate cleanly and *mechanistically*: the payment-panic mouse bids near zero, while the second-price-overgeneralizer mouse bids *above its value* (median ratio 1.15) because it applies the second-price intuition “I will not pay my own bid” to a mechanism where the winner does pay their own bid. In a first-price auction that error is not harmless—it yields negative surplus—giving a concrete, mechanism-level account of a winner’s-curse pathology.

8.2 From engineered deficits to human-error theories

The stress test is most useful if each engineered behavior corresponds to a documented human phenomenon with a theoretical account. Table 6 draws that map. The procedural deficits correspond to failures of contingent reasoning, limited lookahead, and non-equilibrium beliefs; the authored error mice correspond to overbidding/winner’s-curse and to extreme loss aversion. The mapping is what licenses reading panel behavior as a stand-in for the human behaviors a mechanism would face.

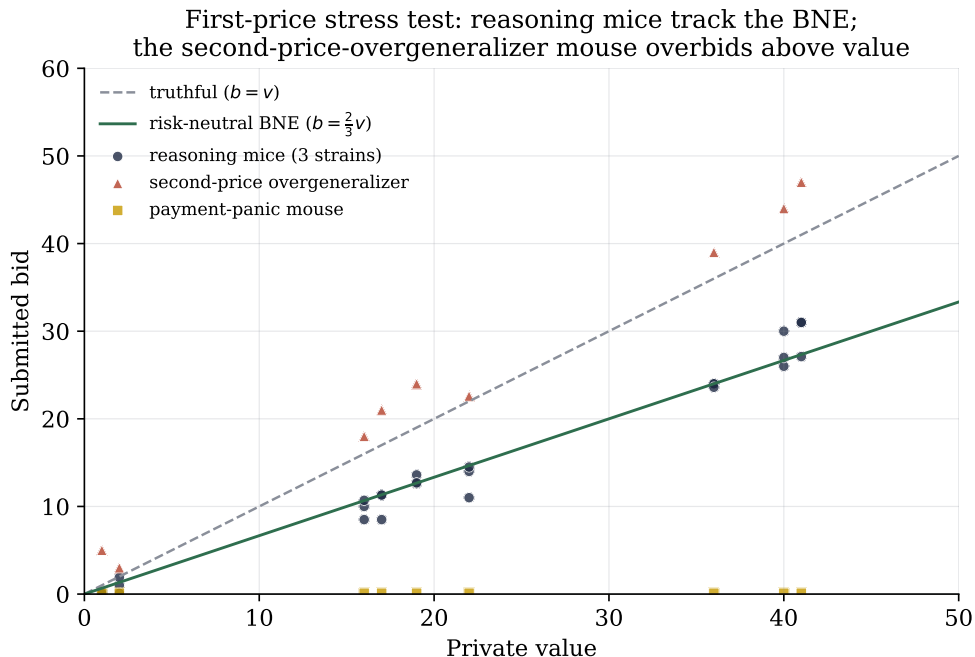


Figure 6: First-price stress test. Submitted bid versus private value for the panel. Reasoning strains cluster on the risk-neutral BNE line $b = \frac{2}{3}v$; the second-price-overgeneralizer mouse bids above the 45° value line (loss-making overbidding); the payment-panic mouse bids near zero.

8.3 Cross-model portability and the strict 5-mouse panel

The first-price result raises a prior question: do the deficit *strains* themselves behave consistently across models? To test this cheaply we reserve a *strict 5-mouse panel*—a small, fixed subset of five strains built on the strict base of Section 2, distinct from the full 27-cell $3 \times 3 \times 3$ basis. We use this five-strain set (rather than all 27) wherever re-running on a second model would otherwise be prohibitive: here for portability, and in Section 9 as the evolutionary search’s evaluator. The five are the two corners (c0_f0_b0, c2_f2_b2) and three reserved mixed strains, chosen to span the behavioral classes while keeping the second-model re-run small.

The pilot that prompted this (Section 7) exposed a failure mode—the stronger model spontaneously repaired the most deficient strain into truthful Vickrey bidding—so the strict scaffold states the reasoning-procedure constraint explicitly rather than as a secondary goal. We then re-ran five reserved strict strains under both base models and compared their dominant behavioral class (Table 7). Four of five preserve their class across models—a partial pass. The exception is the strain isolating a contingent-reasoning deficit, whose moderate-underbidding signature on `gpt-5.4-mini` is repaired into truthful bidding on

Table 5: First-price bid/value ratios by strain ($n = 9$ bidder-decisions each). The risk-neutral BNE benchmark is $\frac{2}{3} \approx 0.667$. Reasoning strains sit on the benchmark; the error mice separate above value and near zero.

Strain	Role	mean b/v	median b/v
c0_f0_b0	reasoning (all-ablated)	0.61	0.65
c2_f0_b2	reasoning (forward-off)	0.61	0.67
c2_f2_b2	reasoning (all-engaged)	0.52	0.66
overgeneralizer	error: overbid	1.61	1.15
payment panic	error: near-zero	0.02	0.01

Notes. Pooled over reasoning strains, the mean ratio is 0.67 (the BNE), 38% of bids exceed the benchmark, and 20% exceed value. The overgeneralizer’s above-value bids are loss-making under first-price pricing.

gpt-5.4 (truthful rate $0.33 \rightarrow 1.00$). Consistent with the construct-validation finding that contingent reasoning is the highest-leverage axis (Table 2) and with the reasoning audit’s finding that it is the leakiest deficit (Section 6), it is also the hardest deficit to hold fixed against a more capable model: the strict scaffold mitigates the portability problem but does not fully close it.

9 Evolutionary Search over Auction Mechanisms

So far the panel has *verified* mechanisms picked by hand. The natural next question is whether the same panel can *drive* a search over mechanisms. This section makes that loop concrete: why a search is the right tool, what the design space is, and how the operators act on an actual auction. We then report the search in *two passes*. Pass 1 is an exploratory, permissive run across mechanism classes (first-price, second-price, third-price, posted-price); it shows the loop works and motivates a sharper question. Pass 2 is a hardened, *scope-locked* run restricted to first-price IPV, with semantic validators that reject malformed or out-of-scope candidates before any digital-mouse evaluation, and a portability check against a stronger model. The two passes serve different ends: the first asks whether the loop discovers sensible mechanism *classes*; the second asks what the loop finds inside a single class once the pipeline is clean enough to trust the ranking.

9.1 Why search, and why language models make it possible

Mechanism design is at bottom a search problem: among the auctions one could implement, which perform well? The standard answer is analytic—Myerson’s optimal auction, Vickrey–

Table 6: From engineered deficits to human-error theories. Each row links a panel strain to its procedural restriction, its SPSB/FPSB signature, and the documented human phenomenon (and theory) it stands in for.

Strain / deficit	Procedural restriction	Signature	Human counterpart (theory)
Contingent ablated ($c=0$)	no opponent-contingent or dominance reasoning	underbids; misses truthful dominance	failure of contingent reasoning; obvious strategy-proofness (Li, 2017)
Forward ablated ($f=0$)	no consequence tracing; margin-of-safety under-bid	underbids ($\approx 60\text{--}80\%$ of value)	limited lookahead / myopia; cognitive hierarchy
Belief ablated ($b=0$)	no model of others	mild miscalibration	non-equilibrium beliefs; level- k (Crawford and Iriberri, 2007), cursed equilibrium (Eyster and Rabin, 2005)
Overgeneralizer (er-ror)	“winner pays second price, so I won’t pay my bid”	overbids above value; loss-making in FPSB	overbidding / winner’s curse (Kagel and Levin, 1993; Crawford and Iriberri, 2007)
Payment panic (er-ror)	minimize the bid one might pay	near-zero bids	extreme loss/risk aversion; participation failure

Clarke–Groves, obvious-strategy-proof variants (Li, 2017; Vickrey, 1961)—each derived in closed form from a tight model of rational bidders. Replace the rational benchmark with a panel of boundedly rational digital mice and the closed-form route closes: “best for the strict 5-mouse panel” is not a function one can differentiate.

An *evolutionary algorithm* is a heuristic search for exactly this situation; it needs only the ability to *evaluate* a candidate. The loop: (i) hold a population of N candidate mechanisms; (ii) score each by a fitness F ; (iii) retain the top performers (*elites*); (iv) form a new generation by perturbing elites (*mutation*) and recombining pairs (*crossover*); (v) iterate. Such searches are routine in machine learning but rare in mechanism design, where candidates have had to be numeric vectors in a fixed family and bidders simulated by a closed-form behavioral model. Frontier language models loosen both constraints: on the bidder side the digital-mouse panel supplies instructable, cognitively varied agents; on the mechanism side an LLM compiles a structured *genotype* (a JSON record of payment rule, bid language, reserve, framing) into a natural-language rule the bidders can act on. Mutation and crossover act on the genotype—“flip `bid_language` from `continuous` to `discrete_grid`” is an unambiguous, schema-validated edit—while the bidder-facing artifact stays language.

Table 7: Strain-level cross-model portability on the strict 5-mouse panel. Dominant behavioral class of five reserved strict strains under each base model. Four of five agree; the contingent-isolated strain’s deficit is repaired away by the stronger model.

Strain	Isolated role	mini → gpt-5.4 (dominant class)	Agree
c2_f2_b0	belief deficit	truthful → truthful	✓
c2_f0_b2	forward deficit	mod. underbid → mod. underbid	✓
c0_f0_b1	near-all-ablated	mod. underbid → mod. underbid	✓
c1_f1_b1	all-intermediate	truthful → truthful	✓
c0_f2_b2	contingent deficit	mod. underbid → truthful	—

Notes. Agreement = 4/5. “Isolated role” names the axis held in its deficit state (others engaged), or a corner strain. The contingent-deficit strain’s truthful rate rises $0.33 \rightarrow 1.00$ across models; all other strains’ dominant class is stable.

What “fitness” means here. F_{robust} (7) is panel-based: a candidate scores well only if it is efficient, induces its declared intended action, and does so *across* the panel, not just for a rational outlier. The search is therefore explicitly looking for mechanisms that are robust to bounded rationality, not for mechanisms that are optimal against a rational benchmark.

9.2 The design space (an integrative-experiment view)

Following integrative experiment design (Almaatouq et al., 2024; Alsobay et al., 2025), we do not search a single knob but a structured space. Table 8 lists the design axes; a genotype (`single_item_auction_genotype_v1`) is one point in this space, augmented by framing fields (description style, scaffold, whether to include a CoT demonstration) and a `target_policy` block declaring the benchmark bid that feeds m_τ (4) and SMAD (1). A concrete top genotype:

```
environment: {value_model: IPV, n_bidders: 3,
  private_range: [0,49], increment: 0.1}
mechanism: {payment_rule: first_price, bid_language: continuous_bid,
  reserve_rule: none, tie_breaking: random}
framing: {description_style: menu, scaffold: none, cot_demo: false}
target_policy: {benchmark: ((n-1)/n)*value,
  target_bid_value_ratio: 0.667, tolerance: 0.1}
```

Table 8 is the *conceptual* target: the full space of mechanism classes and failure modes the panel is meant to screen, in the phase-I spirit of Section 1. The two passes below cover it at two different widths. Pass 1 lets the operators roam across pricing rules to confirm the loop behaves sensibly. Pass 2 then locks the executable search to a single cell of this

Table 8: The single-item-auction design space the search moves over. A genotype is one selection from each axis; the evolutionary operators edit these fields and an LLM rewrites the rule text to match.

Design axis	Values explored
Pricing rule	first-price, second-price, third-price, all-pay, average-price
Payment modifier	none, with-reserve, entry-fee, refundable-deposit
Bid language	continuous, discrete-grid, binary-accept, ranking
Information	private-only, post-clearing revelation, pre-bid signal
Simplicity emphasis	none, low, medium, high

table—single-item sealed-bid, IPV, three bidders, first-price payment, no reserve—so that the rule text the mice actually see is enforceably consistent with the genotype. Narrowing is deliberate: it trades breadth for a ranking we can defend cell by cell, rather than a winner that may hinge on a malformed or ambiguous rule.

It helps to watch the operators act on a real mechanism. Take the rank-2 elite, a *first-price continuous* sealed-bid auction (the gen-0 seed). An *elite* step simply copies it unchanged into the next generation because it is in the top quartile. A *mutation* step resamples one field—say it flips `bid_language` from `continuous` to `discrete_grid`—and asks the LLM to rewrite “submit any bid in \$0.1 increments” into “choose a bid from the grid {0, 1, 5, 10, 20, 30, 50},” producing the rank-3 *first-price discrete-grid* mechanism. A *crossover* step takes two parents—a first-price continuous auction and a mechanism carrying a **decision-table** framing—and, field by field, draws each child field from one parent, yielding for instance a first-price continuous auction that inherits the other parent’s framing scaffold; the LLM then rewrites a single coherent rule. A schema validator rejects ill-typed offspring and duplicates (by rule-text hash) are discarded. The gen-1 crossover child built this way is the eventual rank-1 mechanism. Pass 2 adds a stricter gate in front of this step—a chain of *semantic* validators described in Section 9.5—that rejects out-of-scope or self-inconsistent offspring before any mouse evaluates them.

9.3 Pass 1: an exploratory cross-mechanism search

The run uses 24 individuals for 2 generations against the strict 5-mouse panel with 2 repetitions per mouse—48 mechanism evaluations and 480 raw bidder-decisions, on `gpt-5.4-mini`. It is a proof-of-concept of the EA-plus-panel loop, not a paper-scale search. Figure 7 and Table 9 report two findings. First, the search *improves*: the best-of-generation fitness rises from 3.37 (a gen-0 initialization) to 3.42 (a gen-1 crossover

Evolutionary search over auction genotypes (24 individuals $\times$ 2 generations $\times$ 5-mouse panel)

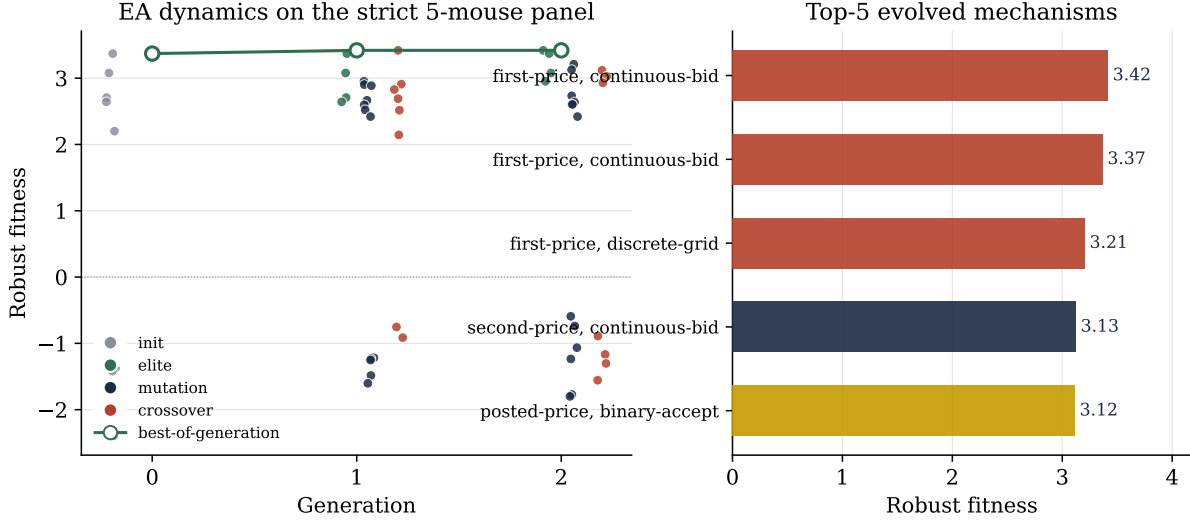


Figure 7: Evolutionary search over single-item auction genotypes against the strict 5-mouse panel (24 individuals $\times$ 2 generations $\times$ 5 personas $\times$ 2 reps = 480 raw bidder-decisions). Left: robust fitness of each individual by generation, colored by operator; the green line traces the best-of-generation. Right: the top-five unique mechanisms, colored by payment rule.

child), which remains elite through gen-2. Second, and more substantively, the top of the population concentrates on canonical sealed-bid designs: three of the top five evolved mechanisms are first-price sealed-bid (continuous or discrete-grid), one is second-price continuous, and one is a posted-price binary-accept design; all five share no reserve and random tie-breaking. The EA, given a permissive genotype space and the digital-mouse panel as evaluator, elevates first-price continuous—exactly the mechanism the a-priori stress test of Section 8 picked—to the top.

9.4 A transparent re-ranking by SMAD

F_{robust} is a multi-objective composite and hence hard to read. As a check we re-rank the evolved mechanisms by SMAD (1), the percentage deviation of realized bids from each mechanism’s *own* declared optimum (Figure 8). The two rankings broadly agree—the Spearman correlation between $-\text{SMAD}$ and F_{robust} is 0.54—but they tell complementary stories, and the disagreement is informative. Second-price mechanisms attain the *lowest* SMAD (the best second-price design reaches $\text{SMAD} \approx 37$, and second-price continuous ≈ 39): under a strategy-proof rule, bidders find the mechanism’s optimum (truthful bidding) most easily. First-price mechanisms—which *win* on F_{robust} —sit higher (≈ 45 – 51),

Table 9: Top-five evolved mechanisms, ordered by robust fitness. Three of five are first-price sealed-bid; all share no reserve and random tie-breaking. The gen-1 crossover child improves on every gen-0 initialization.

Rank	Payment rule	Bid language	Reserve	Framing scaffold	Fitness	First seen
1	first-price	continuous	none	none	3.42	gen 1, crossover
2	first-price	continuous	none	none	3.37	gen 0, init
3	first-price	discrete-grid	none	decision-table	3.21	gen 2, mutation
4	second-price	continuous	none	none	3.13	gen 2, mutation
5	posted-price	binary-accept	none	short-hint	3.12	gen 2, crossover

and third-price is worst (≈ 66). The reading is clean: second-price is the easiest mechanism to bid in *optimally*, while first-price is the mechanism the panel-based robustness objective *prefers*. Both statements are true and they answer different questions, which is exactly why we report both rather than collapsing to one number.

We read the overall agreement as *convergent validation* of two procedures—the a-priori choice of Section 8 rests on sharp theory and human data; the EA choice rests on panel-based robustness; they agree, which they need not have—while the SMAD/fitness split is a reminder of the search’s actual objective. The search is explicitly looking for mechanisms that are robust to bounded rationality, not for mechanisms that minimize bidders’ deviation from a rational optimum; first-price wins the former and second-price the latter.

From Pass 1 to Pass 2. Pass 1 is a proof-of-concept, not a paper-scale search: two generations, calibration-aligned default weights, and—most importantly—a permissive genotype space in which the LLM is free to drift across pricing rules. That permissiveness is what makes Pass 1 a useful sanity check, but it also leaves two soft spots. First, when offspring can carry any pricing rule, the generator occasionally emits a rule text that disagrees with its own genotype (a “first-price” record described to bidders as paying the second-highest bid), and nothing in Pass 1 catches it before the mice bid. Second, the ranking has not been checked against a stronger model, so a winner could be an artifact of the small search model. Pass 2 closes both gaps: it locks the scope to a single first-price cell, adds semantic validators that reject inconsistent or out-of-scope candidates *before* evaluation, and re-scores survivors on a stronger held-out model.

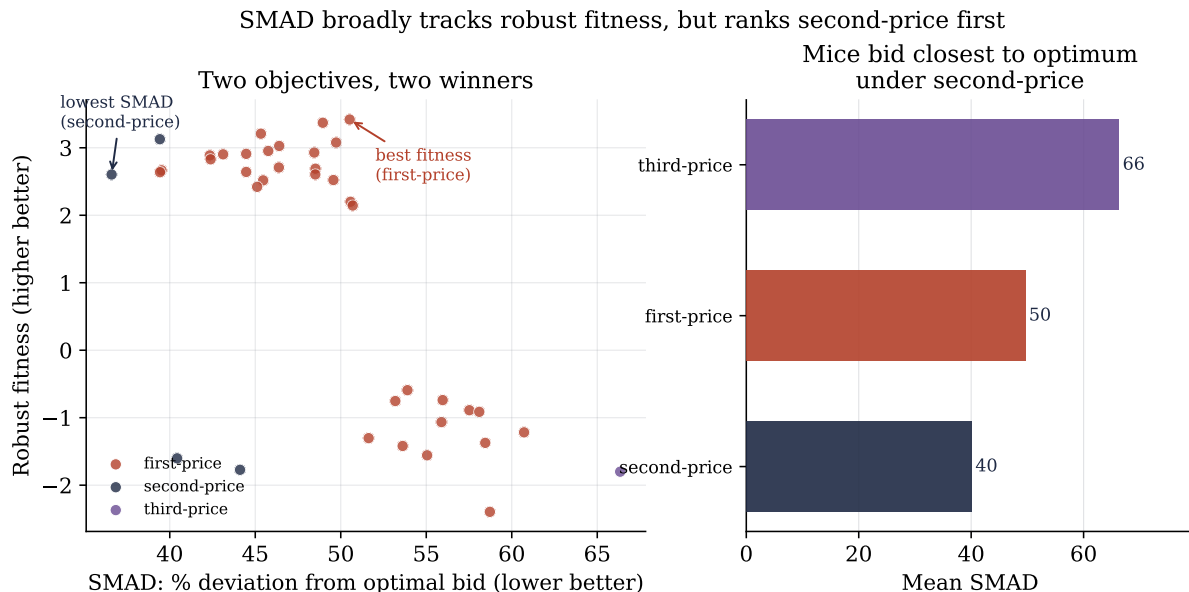


Figure 8: SMAD as a transparent cross-check on robust fitness. Left: robust fitness versus SMAD (scaled deviation from each mechanism’s own optimum) for the evolved mechanisms, colored by payment rule; the two best-by-each-criterion mechanisms are annotated. Right: mean SMAD by payment rule—second-price is lowest (easiest to bid in optimally), third-price highest.

9.5 Pass 2: a hardened, scope-locked first-price search

Pass 2 asks a sharper question than Pass 1: holding the mechanism *class* fixed at first-price IPV, what does the panel-driven search prefer *within* that class, once the pipeline is clean enough that every ranked candidate is a genuine, well-formed first-price auction? Fixing the class is what makes the question answerable—differences in the ranking can no longer be confounded by a candidate having silently become a different mechanism.

Scope lock and semantic validators. A genotype-scope constraint (`first_price_ipv`) restricts the executable search to the target cell: IPV with three bidders, sealed-bid, highest-bid-wins, first-price payment, no reserve, no binary accept/reject. Mutation and crossover may still edit the *executable* first-price dimensions—bid language, description style, scaffold, feedback—but cannot mutate into a different payment rule or value environment. On top of the scope lock, a chain of semantic validators screens every candidate *before* it reaches a mouse: a phenotype-consistency check that the generated rule text agrees with the genotype; a strict dry-render that catches unresolved template variables before the text is shown to bidders; a payment-rule detector that rejects rule text without a clear first-price cue or with second-/third-price wording presented as the actual rule; and

an action-interface check that the rule exposes a numeric-bid action (rejecting binary accept/reject in this version). Crucially, these validators police the *mechanism*, not the *mouse*: a bounded-rationality strain that *believes* “the winner pays the second-highest bid” while playing a first-price auction is a measured cognitive failure and is preserved, not filtered.

The run. Under this hardened pipeline we ran 24 individuals for 2 generations on `gpt-5.4-mini` (seed 20260531), scored against the same strict 5-mouse panel with 2 repetitions per mouse. Strict validation and a fraction of remote-inference failures mean not every candidate slot completes: the run yielded 44 valid, fully scored mechanisms (8 in gen-0, 17 in gen-1, 19 in gen-2) from 440 raw evaluated mouse decisions. We treat these 44 as a diagnostic population, not a final leaderboard—the completion rate is itself one of the findings.

What the search prefers. Figure 9 and Table 10 report the result. The scope lock held: no evaluated mechanism drifted into second-price, third-price, posted-price, all-pay, common-value, or binary accept/reject; candidates that tried to drift were rejected before evaluation. Within the clean first-price neighborhood, the top of the ranking concentrates on *grid and scaffolded* first-price designs rather than the continuous-bid design that led Pass 1—the best is a short-hinted first-price sealed-bid grid at $F_{\text{robust}} = 3.23$. Two features of the population explain the ranking. Allocative efficiency is uniformly high (mostly ≥ 0.95), so it does *not* separate good from bad candidates; what discriminates is the *worst-persona* lower tail, which collapses toward 0.1 for the negative-fitness mechanisms while the mean stays high (Figure 9, right). Robust fitness here is almost entirely a story about the lower tail, not the average—exactly the behavior the tail-adjusted objective (7) was built to reward.

Portability against a stronger model. A high score on the small search model is only worth keeping if it survives a stronger one. We therefore re-score survivors on a stronger held-out model and report a *portability-adjusted* fitness, $F_{\text{port}} = F_{\text{mini}} - \max(0, F_{\text{mini}} - F_{\text{full}})$, which equals the smaller of the two robust fitnesses and so demotes any candidate that collapses under the stronger model. Most leaders are stable—a BNE-hinted first-price design scores 3.21 on the search model and 3.27 on the stronger model (no penalty), and first-price continuous moves $3.07 \rightarrow 3.34$ (no penalty). The check earns its keep on the failures: a coarse-grid first-price design that looks strong on the search model ($F_{\text{mini}} = 3.07$) collapses on the stronger model ($F_{\text{full}} = -1.29$), a penalty of 4.36 that

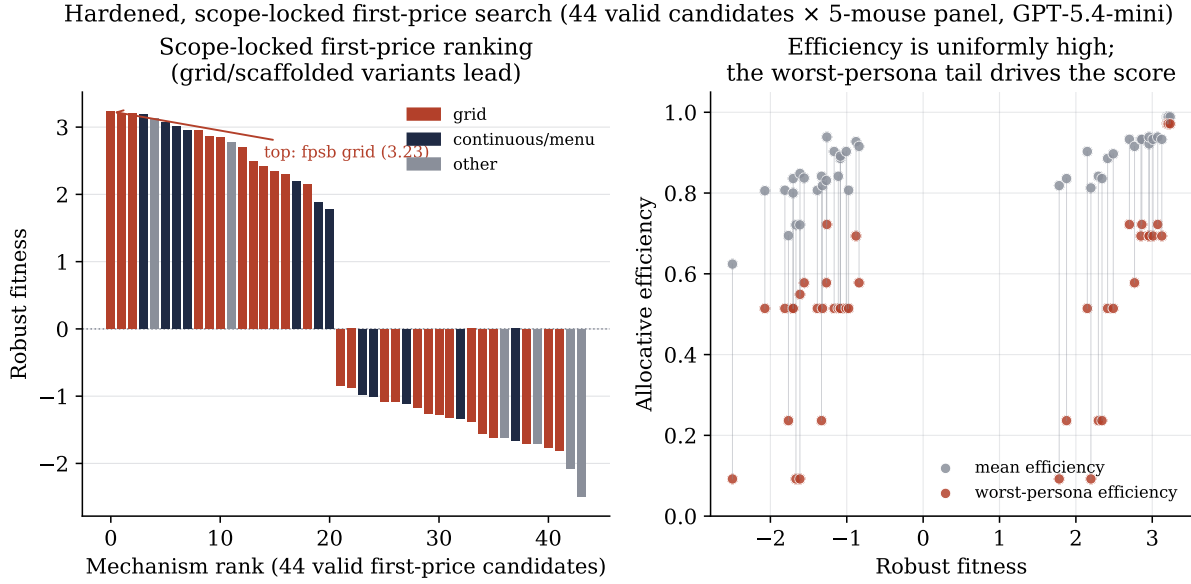


Figure 9: Hardened, scope-locked first-price search (44 valid candidates against the strict 5-mouse panel, `gpt-5.4-mini`). Left: the 44 candidates ranked by robust fitness and colored by bid-language family; grid and scaffolded variants lead, and a long negative tail of mechanisms is penalized. Right: mean allocative efficiency (grey) stays uniformly high across the population, while the worst-persona efficiency (red) fans down for low-fitness mechanisms—the lower tail, not the mean, drives the score.

pushes its portability-adjusted fitness to -1.29 . Coarse grids, it turns out, work only because the smaller model coordinates on a convenient grid point; the stronger model does not, and the design’s apparent robustness evaporates.

Caveats, and why we have not scaled yet. Three limits keep this a diagnostic pass. (i) Action-space enforcement is still only at the rule-text level: the validators check that a grid is *described*, but the current runner and scorer would accept any numeric bid, so grid-versus-continuous revenue and simplicity differences are not yet enforced where they are measured. (ii) Completion is partial (44 of the 72 candidate slots), a mix of strict-validator rejections and remote failures, so the ranking is a guide, not a verdict. (iii) The grid leaders are promising but not yet cleanly comparable to continuous designs until enforcement moves into the runner. For these reasons we deliberately did *not* launch a larger EA on this result. The next engineering step is runner/scorer-level action-interface enforcement and an explicit action-compliance penalty, after which the same first-price scope can be rerun at scale and only the finalists carried to a full-model portability test.

Table 10: Top-five mechanisms from the hardened, scope-locked first-price search, ordered by robust fitness. All are first-price by construction; the leaders are grid or menu/scaffolded variants. “Intended-action rate” is the share of bids landing in the declared target band; it is low and roughly flat across the leaders, so it is not what separates them—the worst-persona efficiency tail is.

Rank	Mechanism (bid language)	F_{robust}	Intended-action	Revenue	Efficiency
1	fpsb grid, short hint	3.23	0.33	27.10	0.99
2	first-price grid	3.21	0.27	27.90	0.99
3	first-price discrete grid	3.20	0.37	28.50	0.99
4	fpsb continuous menu	3.19	0.37	26.27	0.99
5	first-price sealed-bid	3.13	0.37	28.12	0.93

10 Discussion

Procedural basis versus authored error mice. Construct validity is strongest for the procedural basis, where a single instruction (engage an axis) produces the predicted directional movement. Calibration is more nuanced: the basis covers truthful and underbidding behavior on its own, but overbidding—a first-order regularity in human data (Kagel and Levin, 1993)—is recovered only with hand-specified heuristic strains. The model-agnostic claim applies cleanly to the procedural axes; the error mice are a complementary, more bespoke device for spanning the mistake space, and the two should be reported as distinct epistemic objects.

Digital twins as funhouse mirrors. Our design is a direct response to the distortions Peng et al. (2025) document in LLM digital twins. We do not attempt the thing those distortions make unreliable—reproducing the full distribution of human opinions or behaviors with a single model—because that is both unattainable and heavily model-dependent. Instead we exploit one distortion, hyper-rationality, as a clean substrate to engineer *specific* deficits from, and we measure model-dependence directly rather than assuming it away (Section 8). The reasoning audit (Section 6) is in the same spirit: it is an internal-validity check that the induced deficit is the one we intended, not an emergent artifact.

Model dependence. Because deficits are procedural controls, the protocol can be rerun on a different frontier model. At the strain level, four of five reserved strict strains preserve their dominant class across base models; the lone exception is the contingent-reasoning deficit, which a stronger model tends to repair into truthful bidding. This is a feature, not a bug—the panel is the instrument that makes such repair visible—but it cautions against reading single-model panel scores as model-invariant and identifies contingent reasoning as

the deficit most in need of a hardened scaffold; the demonstration result (Section 7) is one remedy.

External validity. Digital mice are a phase-I screening population. A strain reproducing a mistake class establishes that the panel can express that behavior, not that the human share of that behavior is matched. We therefore keep the stress-testing claim (this paper) separate from any calibrated prediction of human behavior, which would require human pilot data—the subject of the next section.

11 Toward Validation with Human Experiments

The pipeline so far ends in a short list of candidate mechanisms; it does not end in a human-validated one. The final link is to test the panel’s verdicts against people, and the integrative-experiment framing makes the test well-defined rather than ad hoc (Almaatouq et al., 2024; Alsobay et al., 2025). The design space of Table 8 is exactly the pre-committed family over which one can make externally valid statements: draw a random sample of mechanisms from it, assign them to human bidders, and compare two things. First, *behavioral* validation—do human bidders in a sampled mechanism produce the bid distribution the digital-mouse panel predicted for it (truthful core plus the documented overbidding tail in first-price, etc.)? Second, *ranking* validation—does the panel’s robustness ordering over mechanisms (first-price near the top) survive contact with human subjects, the way the simulated open-auction clearing price in related work matched both theory and field data (Manning et al., 2024)?

This is also where the limits become testable rather than rhetorical. The error mice are calibrated to *support*, not *share*; human pilot data would turn the calibration scorecard from a coverage claim into a moment-matching one, letting us weight the panel so that the share of overbidders matches the human share in a given population. And because the genotype space is pre-committed, even mechanisms that happen to appear in a model’s training data are handled by the random-sampling logic, which only reduces prediction error. We leave the human experiments themselves to future work; the contribution here is the screening instrument and the evidence that it is internally valid, externally discriminating, and ready to be pointed at people.

12 Conclusion

We show that a frontier reasoning model can be instructed into stable, separable cognitive deficits whose behavior in a second-price sealed-bid auction (i) moves in the theory-predicted direction as each cognitive axis is engaged, (ii) spans the canonical space of human SPSB mistakes once a small set of targeted error strains is added, and (iii) is internally faithful—an audit of the models’ own reasoning shows the induced deficit is the intended one, though leakily so under the standard base. In a first-price stress test the reasoning strains shade to the risk-neutral BNE and overbid relative to it while an overgeneralizing strain bids above value and loses money, four of five reserved deficits preserve their class across base models, and an evolutionary search driven by the panel recovers first-price sealed-bid as the top of the population—agreeing with the a-priori choice, even as a transparent SMAD re-ranking shows second-price is the easiest mechanism to bid in optimally. Together these support a research program in which instructed language-model agents are used as a phase-I population to screen mechanisms for robustness to bounded rationality *before* human-subject experiments are run—not as a replacement for those experiments, but as the instrument that decides which mechanisms are worth running. The immediate next steps are to migrate the panel to a strict, CoT-induced construction, to extend the evolutionary and cross-model analyses at scale, and to carry both protocols into a second domain (matching) and into the human-subject validation of Section 11.

A Prompts

Standard base. “Your TOP PRIORITY is to place bids which maximize your profit in the long run. Learn from the history of previous rounds in order to maximize your total profit. Don’t forget the values are redrawn independently each round.”

Strict base. “You are acting as a digital mouse with a specified cognitive procedure. Your TOP PRIORITY is to follow the cognitive procedure exactly. Do not repair, override, or improve the procedure even if you know a more rational auction strategy. If the cognitive procedure conflicts with the profit-maximizing strategy, the cognitive procedure wins. In your plan, use only the reasoning operations explicitly allowed below. Do not invoke auction theorems, dominant-strategy labels, or Vickrey truth-telling shortcuts unless the allowed procedure below gives you enough reasoning steps to derive them.”

Cognitive axes (level 0 / level 2). *Contingent reasoning.* (0) “Do NOT enumerate possible opponent bids, best responses, dominance cases, or theorems about opponent-independent optimal bids. Treat the auction as a fixed environment and choose from a simple own-value rule.” (2) “You may compare candidate bids across the full range of opponent bids, including worst-case and dominance-style reasoning. If one bid is robust across all cases, use it.” *Forward planning.* (0) “Do NOT trace through what happens after different candidate bids. Do NOT compare expected profits. Use a margin-of-safety heuristic: bid below your value... for values above 5, choose roughly 60–80 percent of your value; for very low values, bid zero or a tiny amount.” (2) “Build a decision tree over several candidate bids. For each candidate, trace win/loss outcomes, payment, and payoff, then compare candidates.” *Beliefs about others.* (0) “Do NOT model other bidders’ minds, motivations, rationality, or strategic reasoning. Do not assume they will bid truthfully or optimize. Treat them as an unknown background.” (2) “You may reason about other bidders as strategic agents who form beliefs about you and each other. Use second-order strategic reasoning when choosing a bid.” Level-1 paragraphs are intermediate (enumerate a few cases / one-step lookahead / first-order beliefs).

Error mice. *Second-price overgeneralizer:* “You strongly focus on the fact that the winner pays the second-highest bid, not their own bid. Overgeneralize this rule: treat bidding above your value as mostly safe because you expect not to pay your own bid.” *Payment panic:* “You are afraid that your bid is the amount you might have to pay... choose the smallest positive bid allowed by the auction grid... Even if your value is high, protect yourself by bidding 0.1 rather than trying to win.”

CoT demonstration (Section 7). A single worked example is appended after the cognitive-style block and before the live decision. It fixes a sample value, walks through only the allowed reasoning, and ends in a bid: a margin-of-safety underbid ($\approx 0.7 \times$ value) for the all-ablated strain, and the contingent/forward/belief reasoning that yields a truthful Vickrey bid for the all-engaged strain.

B A Planned Outcome-Instruction Robustness Check

The panel induces deficits by constraining the reasoning *procedure*. A complementary design, which we specify here and leave to a (cost-bearing) re-run, induces them by naming the *outcome* directly: rather than telling a mouse *how* to think, tell it *what* it

does—“you tend to overbid,” “you tend to underbid,” “you bid your value.” Comparing outcome-instructed strains to the procedure-instructed strains of this paper would separate two explanations for the observed signatures: genuine procedural induction versus simple compliance with a labeled behavior. The prediction is that procedure-instructed strains should match their outcome-instructed counterparts on the targeted metric but differ on *off-target* behavior and on the reasoning audit of Section 6 (an outcome instruction need not, and likely will not, change the dominance-argument invocation rate). We flag this as the cleanest single robustness experiment to run next.

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